

7.5 Electrical Characteristics (continued)

Typical conditions are measured at 25°C with nominal BAT voltages of 18 V (BQ76920), 36 V (BQ76930), or 48 V (BQ76940) with $V_{CELL} = 4$ V. Min and max values include full recommended operating condition temperature range from -40°C to +85°C. Certain characteristics may be shown at different voltage or temperature ranges, as clarified in the Test Condition sections.

PARAMETER		TEST CONDITION	MIN	TYP	MAX	UNIT
INTERNAL POWER CONTROL (STARTUP and SHUTDOWN)						
V _{PORA}	Analog POR threshold	V _{BAT} rising. See (4).	4		5	V
V _{SHUT}	Shutdown voltage	V _{BAT} falling. See (4).			3.6	V
t _{I2CSTARTUP}	Time delay after boot signal on TS1 before I ² C communications allowed	Delay after boot sequence when I ² C communication is allowed		1		ms
t _{BOOTREADY}	Device boot startup delay	Delay after boot signal when device has completed full boot-up sequence			10	ms
T _{SHUTD}	Thermal shutdown voltage			100	150	°C
MEASUREMENT SCHEDULE						
t _{VCELL}	Cell voltage measurement interval	BQ76920, BQ76930, BQ76940		250		ms
t _{INDCELL}	Individual cell measurement time	Per cell, balancing off		50		
		Per cell, balancing on		12.5		
t _{CB_RELAX}	Cell balancing relaxation time before cell voltage measured			12.5		
t _{TEMP_DEC}	Temperature measurement decimation time	Measurement duration for temperature reading		12.5		
t _{BAT}	Pack voltage calculation interval			250		
t _{TEMP}	Temperature measurement interval	Period of measurement of either TS1/TS2/TS3 or internal die temp		2		s
14-BIT ADC FOR CELL VOLTAGE AND TEMPERATURE MEASUREMENT						
ADC _{RANGE}	ADC measurement recommend operation range	V _{CELL} measurements	2		5	V
		TS/Temp measurements	0.3		3	V
ADC _{LSB}	ADC LSB value			382		μV
ADC	ADC cell voltage accuracy at 25°C	V _{CELL} = 3.6 V - 4.3 V		±10		mV
		V _{CELL} = 3.2 V - 4.6 V		±15		
		V _{CELL} = 2.0 V - 5.0 V		±25		
	ADC cell voltage accuracy 0°C to 60°C	V _{CELL} = 3.6 V - 4.3 V		±20		
		V _{CELL} = 3.2 V - 4.6 V		±25		
		V _{CELL} = 2.0 V - 5.0 V		±35		
	ADC cell voltage accuracy - 40°C to 85°C	V _{CELL} = 3.6 V - 4.3 V	- 40		40	
		V _{CELL} = 3.2 V - 4.6 V	- 40		40	
		V _{CELL} = 2.0 V - 5.0 V	- 50		50	
16-BIT CC FOR PACK CURRENT MEASUREMENT						
CC _{RANGE}	CC input voltage range		- 200		200	mV
CC _{FSR}	CC full scale range		- 270		270	mV
CC _{LSB}	CC LSB value	CC running constantly		8.44		μV
t _{CCREAD}	Conversion time	Single conversion		250		ms
CC _{INL}	Integral nonlinearity	16-bit, best fit over input voltage range ± 200 mV		± 2	± 40	LSB
CC _{OFFSET}	Offset error			± 1	± 3	LSB